Johnson Jasson

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Summary

Electrical & Electronics Engineering and Mathematics double-major sophomore (4.0 GPA) with hands-on experience in circuit design, embedded systems, power distribution, Python and AI/ML. Interested in building intelligent hardware-software systems with microcontrollers, sensors, and real-time firmware.

Education

Alfred University Expected May 2029

Bachelor of Science in Electrical and Electronics Engineering, Double Major in Mathematics GPA 4.00/4.00

Relevant Coursework: Digital logic, Emerging Nanoelectronics, Machine Learning Applications, Computer Aided Design, Calculus, Data structures and Algorithms, Physics, Circuit Theory.

Cornell University - Break Through Tech Program March 2026 – March 2027

Certificate in Machine Learning Foundations and Artificial Intelligence CGPA: 96.87/100

Technical Skills

Software: Python, Verilog/VHDL (basic), LINUX, Git/GitHub, NumPy, Pandas.
Hardware: ESP32, Arduino, Oscilloscope, multimeter, CMOS, Digital Logic, Soldering.
Engineering Tools: MATLAB, Solidworks, Auto-CAD, Microsoft Excel, LaTex, KiCad (Basic).
AI-Assisted Development: Claude Code, GitHub Copilot, Cursor, Codex (agentic coding workflows).

Technical Experience

Modeling and Simulations Intern May 2026 - July 2026

Undergraduate Summer Research Alfred University

- Modeled polarization switching across 5,000 grains in 8.3 nm polycrystalline Hafnium Oxide FeFET transistor using Python and MATLAB.
- Simulated Set/Reset pulses mapping polarization +22.9 to -22.9 $\mu C/cm^2$ to enable multi-level neuromorphic memory.

Computational Semiconductor Research Assistant January 2026 - May 2026

Knepp Lab, Alfred University Alfred, NY

- Quantified electron coupling across 300–700 K in organic semiconductors via DFT to guide charge-transport design.
- Executed 40+ HPC jobs (60+ core-hours) on Linux, optimizing data pipelines for reproducible simulation results.

Electrical Engineering Intern July 2024 - September 2024

Tanzania National Electric Supply Company (TANESCO) Dar es Salaam, TZ

- Troubleshot a 33 kV transformer fault on a 50 Hz network and supported 230 V service-line and meter installs.
- Raised plant power factor from 0.80 to 0.95 via capacitor-bank fix, cutting monthly electricity bills falling by about 60%, from TZS 5M to nearly TZS 2M.

Projects

Edge Vision Sensing Platform Ongoing

C/C++, ESP-IDF, RTOS, Sensors Alfred, NY

- Built an ESP32 mixed-signal platform integrating IR, PIR, and camera subsystems over validated GPIO interfaces.
- Engineered C/C++ FreeRTOS firmware with interrupt-driven events and queue-based logging for real-time operation.
- Debugged GPIO states, UART, and sensor thresholds across Wokwi simulation and breadboard prototyping.

ESP32 Wi-Fi Smart Switch December 2025

Embedded C++, Arduino IDE, Internet of Things (IoT) Alfred, NY

- Prototyped an ESP32 Wi-Fi switch for remote load control, cutting manual switching time by up to 10 seconds.
- Built scalable firmware with an embedded HTTP server for real-time GPIO control and wireless device management.

SERA, Co-Founder & Lead Developer January 2026 - Present

Full-Stack, AI Agentic Development Alfred, NY

- Architected and shipped a full-stack platform centralizing opportunity applications for students across East Africa.
- Accelerated delivery using AI agentic coding tools (Claude Code, Cursor), deploying a scalable web app on Vercel.
- Drove adoption by leading an Ed Tech competition: 44 students, 11 project submissions, and a $400 innovation prize.

Leadership and Involvement

Robotics Club Alfred University - Vice President January 2026 - Present

- Co-directing operations and member growth while leading a team to build a robotics prototype end to end.

ColorStack Alfred University - Professional Development Chair August 2026 - Present

- Leading professional development for nearly 100 members through weekly resume, interview, and career sessions.

NVIDIA Summer Bridge Series June 2026 - August 2026

- Engaged NVIDIA engineers on GPU computing and AI infrastructure while refining technical interview readiness.